BUSINESS REQUIREMENTS DOCUMENT

QR-Based Visitor Management System

Replacing the Physical Visitor Register at Office HQ and Branches

Field	Details
Document Version	v1.0
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Date	April 7, 2026
Project Type	Internship Project
Document Status	Draft — Pending Review
Reviewed By	MP- Online

1. Executive Summary

This document describes the full set of business requirements for a QR-Based Visitor Management System. The goal is simple: replace the old paper register at the company's front desk with a smart, digital system that makes visitor check-in fast, secure, and easy to track.

Right now, every visitor who walks into the office has to write their name, contact number, and purpose of visit in a physical register. This process is slow, prone to errors, and gives the security team no way to see in real time who is inside the building. There is also no easy way to pull up historical visitor data or generate reports.

The proposed system will fix all of this. Visitors will register on their phone, receive a QR code, and simply scan it at the security desk. The security team will have a live dashboard. Management will be able to pull reports at any time. And as an optional future enhancement, AI features like facial recognition and suspicious pattern detection can be layered on top.

2. Problem Statement

To understand why this system is needed, it helps to walk through what happens today when a visitor arrives at the office:

- The visitor walks up to reception and asks to meet someone.
- The receptionist or security guard hands them a paper register.
- The visitor writes their name, phone number, purpose, and the name of the person they are meeting.
- The guard manually checks the entry and allows or denies access.
- At the end of the day, all this data sits in a paper book — unsearchable, easy to lose, and impossible to anlyse.

This process has several clear problems. There is no way to know in real time how many people are inside the building. There is no automatic notification to the host employee. Entries can be forged or left incomplete. And pulling any historical report requires someone to manually go through the register.

The QR-Based Visitor Management System addresses all of these problems directly.

3. Project Scope

3.1 What This Project Will Do

- Allow visitors to register on a mobile-friendly web form before or during their visit.
- Automatically generate a unique QR code for each registered visitor.
- Let security staff scan the QR code at the entry desk using a camera-equipped device.
- Show a real-time dashboard to admins and management with live visitor data.
- Store all visitor records securely and make them available for reporting.
- Support multiple office branches under a single system.

3.2 What This Project Will NOT Do

- Control physical door locks, turnstiles, or any hardware access systems.
- Integrate with HR, payroll, or employee management systems.
- Replace or duplicate the company's existing CCTV or surveillance infrastructure.
- Handle visitor payments or any financial transactions.

4. Who Is Involved — Roles and Responsibilities

There are five main groups of people who will interact with this system. Each has a different role and a different set of things they need to be able to do.

Role	Who They Are	What They Do in the System
Visitor	Any guest, vendor, client, or contractor visiting the office	Fills out the registration form, receives and presents the QR code at entry and exit.
Security Guard	Personnel stationed at the entry gate or reception desk	Scans QR codes to verify visitors, confirms check-ins and check-outs, flags unauthorized entries.
Office Admin / Receptionist	Front desk or admin staff managing day-to-day operations	Handles walk-in registrations, manages the dashboard, exports reports, assists visitors who do not have a smartphone.
System Administrator (IT)	The IT team member responsible for the system	Manages user accounts and roles, configures system settings, handles multi-branch setup, oversees backups and data retention.

5. What the System Must Do — Functional Requirements

This section lists everything the system needs to be able to do. Each requirement is given a unique ID and a priority level. High priority items are core to the system — it cannot go live without them. Medium priority items are important but can follow in a second release. Low and Future priority items are enhancements.

5.1 Core Requirements (Must Have)

FR-01: Visitor Registration

The system must provide a simple, mobile-friendly web form that any visitor can fill out on their phone. The form should collect the visitor's full name, contact number, a photo or ID number, the purpose of their visit, and the name of the employee they are here to meet. The form should be quick to fill — under two minutes.

FR-02: QR Code Generation

As soon as a visitor completes their registration, the system must automatically generate a unique QR code for them. This QR code should appear on the visitor's screen immediately and also be sent to their email address. The QR code must be linked to that specific visit only and must expire after a set time (such as 8 hours).

FR-03: Security Desk Check-In

The security guard's interface must allow them to scan a visitor's QR code using a tablet or computer with a camera. When scanned, the visitor's details must appear on screen instantly. The system must log the exact time of check-in automatically. If a QR code has expired or been used before, the system must alert the security guard.

FR-04: Check-Out Logging

When a visitor leaves, the security guard must be able to scan their QR code again to log their departure. The system should automatically calculate and record the total duration of the visit. This

helps management understand how long visitors typically stay and ensures accurate occupancy data.

FR-05: Real-Time Admin Dashboard

The admin dashboard must show, at any given moment, exactly who is currently inside the building, how many visitors have come in today, and a list of all recent check-ins and check-outs. The dashboard should refresh automatically every 30 seconds so that the information is always current.

5.2 Important Requirements (Should Have)

FR-08: Multi-Branch Support

The system must support multiple office locations. Each branch should have its own dedicated dashboard and entry logs. A central administrator should be able to view data across all branches from a single login. QR codes should be location-specific so they cannot be used at a different branch.

FR-09: Visitor Reports and Analytics

The system must allow admins to generate reports. These reports should cover daily, weekly, and monthly visitor counts, the busiest times of day, which departments receive the most visitors, and the average visit duration. Reports must be exportable as PDF or Excel files so they can be shared with management.

5.3 AI-Powered Enhancements (Future Phase)

FR-10: Face Authentication[Optional]

In a future phase, the system could use facial recognition technology to verify visitors at entry. A camera at the security desk would scan the visitor's face and match it against their registration

photo. For frequent or pre-approved visitors, this would allow entry without needing to show a QR code at all. This feature would use OpenCV or a cloud-based service like AWS Rekognition.

6. How Well the System Must Perform — Non-Functional Requirements

Beyond what the system does, we also need to define how well it does it. The following requirements set the standards for performance, security, reliability, and usability.

6.1 Speed and Performance

- Scanning a QR code and completing a check-in must happen within 2 seconds.
- The admin dashboard must fully load within 3 seconds.
- The system must be able to handle at least 100 people using it at the same time without slowing down.
- During peak hours, the system should be able to process at least 50 visitor check-ins per hour without any delays.

6.2 Security

- All QR codes must be time-limited (valid for 8 hours) and single-use. Once a QR code has been scanned for entry, it cannot be reused.
- All visitor data must be transmitted over HTTPS and stored using AES-256 encryption.
- The system must use Role-Based Access Control (RBAC). This means a security guard can only see what a security guard needs to see, an admin can see more, and an IT administrator has full access.
- Every action taken in the system — logins, check-ins, data exports — must be recorded in an audit log that is kept for at least one year.

6.3 Availability and Reliability

- The system must be available 99.5% of the time during office hours (Monday to Saturday, 8 AM to 8 PM).
- A basic offline mode must allow security guards to continue checking in visitors even if the internet goes down temporarily. Data should sync automatically once the connection is restored.

- The database must be backed up automatically every night.
- If something goes wrong, the system must be back up and running within 2 hours (Recovery Time Objective).

6.4 Ease of Use

- A security guard with no prior training should be able to learn how to use the system in under 30 minutes.
- The visitor registration form must work smoothly on any modern Android or iOS phone.
- The interface should be available in both English and Hindi so that all staff and visitors can use it comfortably.
- The design must follow accessibility guidelines (WCAG 2.1) so that it works for users with visual or motor impairments.

6.5 Scalability

- The system must be designed so that new office branches can be added without any significant rework to the codebase or infrastructure.
- The system should be built using an API-first approach, so that future integrations — for example, with access control hardware or an HR system — are straightforward.

6.6 Data Privacy

- Visitor data must be automatically deleted after 90 days, in line with data minimization principles.
- The system must comply with the Information Technology Act, 2000 (India).
- Visitors must give digital consent before their data is collected. The registration form must include a brief, plain-language explanation of how their data will be used.
- Visitor data will never be shared with any third party.

7. Technology Stack

The following table outlines the key technologies that will be used to build this system. These have been chosen to match the skills typically developed during a software internship and are well-supported by open-source libraries.

Layer	Technology	Why This Was Chosen
Frontend (User Interface)	React.js	Industry-standard JavaScript framework — fast, component-based, and ideal for building the visitor form, security scan screen, and admin dashboard.
QR Code Scanner	React-QR-Reader	A simple React library that accesses the device camera and reads QR codes in real time. Works on desktop and mobile browsers.
QR Code Generator	python-qrcode (Python)	A well-known Python library that generates QR code images. Will run on the backend as part of the registration API.
Backend (Server)	Python with FastAPI	Python is fast to develop in and pairs naturally with the QR generation library. FastAPI is preferred for its automatic API documentation.
Database	PostgreSQL	A reliable, production-grade relational database that handles structured visitor data, user roles, and logs very well.
Authentication	JWT Tokens + RBAC	JSON Web Tokens are a standard way to handle secure logins. Combined with role-based access control, they keep each user limited to what they need.
Notifications	SMTP Email / Twilio SMS	Standard tools for sending automated email and SMS notifications to host employees when a visitor checks in.

AI (Face Auth) — Future	OpenCV / DeepFace/ InsightFace /FaceNet,	OpenCV for local face detection

8. User Stories

User stories describe what each type of user needs to be able to do, written from their perspective. They help the development team understand the real-world context behind each feature.

ID	As a...	I need to be able to...	Because...
US-01	Walk-in visitor	fill out a registration form on my phone at reception	I need to be registered in the system before I can enter the office.
US-02	Visitor (any type)	receive my QR code on screen and by email	I need to be able to show it quickly at the security desk without searching through my inbox.
US-03	Security guard	scan a visitor's QR code and instantly see their details	I need to verify who they are quickly without typing anything manually.
US-04	Security guard	be alerted when a QR code is expired or already used	I need to catch fraudulent or reused passes at the gate.
US-07	Office admin	see a live dashboard of who is currently in the building	I need to know the real-time occupancy at any given moment.
US-08	Office admin	export a monthly visitor report as a PDF or Excel file	I need to share this data with management in a format they can work with.
US-09	IT administrator	add a new branch to the system	The company is opening new locations and I need them to be set up quickly.
US-10	IT administrator	view audit logs of all system actions	I need to be able to investigate any security incidents or unusual activity.

9. Risks and How We Plan to Handle Them

Every project has risks. The table below lists the most likely risks for this system and what we plan to do about each one before it becomes a problem.

Risk	How Likely	How Serious	What We Will Do About It
A visitor does not have a smartphone	Medium	Medium	Provide a fallback — the receptionist can complete registration on behalf of the visitor using the admin interface. A kiosk-style tablet at reception also solves this.
Internet connection goes down at the office	Low-Medium	High	Build an offline mode that allows basic check-in to continue locally. Data syncs to the server automatically once connectivity is restored.
A QR code is copied, shared, or forged	Low	High	Make all QR codes time-limited (30 days) and monthly-use. The system checks the token validity on every scan and rejects any code that has already been used.
Security staff find the system too difficult to use	Medium	High	Design the security guard interface with one clear action — point camera at QR, see result. Conduct a 30-minute hands-on training session before go-live.
Visitors raise concerns about face recognition (AI phase)	High	Medium	Make facial recognition strictly opt-in. Collect explicit consent. Store face data encrypted. Allow visitors to request deletion of their biometric data at any time.
Data breach or unauthorized access to visitor records	Low	Very High	Enforce HTTPS everywhere, AES-256 encryption at rest, JWT-based authentication, RBAC, and automatic audit logging. Conduct a security review before launch.

10. Acceptance Criteria

The system will be considered complete and ready for handover when all of the following conditions are met during User Acceptance Testing (UAT):

- A visitor can complete the registration form in under 2 minutes on a standard smartphone.
- A QR code is generated and visible on screen within 5 seconds of completing registration.
- A security guard can complete a full check-in by scanning a QR code in under 2 seconds.
- The admin dashboard reflects the latest check-in data within 30 seconds.
- The system correctly rejects any QR code that has expired or been used before.
- The system handles 100 simultaneous users without any errors or slowdowns.
- All data is transmitted over HTTPS. No visitor data is accessible without a valid login.
- Each user role can only access the parts of the system they are permitted to use.
- Monthly visitor reports can be generated and exported in both PDF and Excel formats.
- The system achieves 99.5% uptime during a 2-week UAT period.

11. Assumptions and Constraints

11.1 Assumptions

The following assumptions have been made while writing this document. If any of these turn out to be incorrect, the requirements may need to be updated.

- The office has a stable internet connection during working hours.
- The security desk has at least one tablet or desktop computer with a working camera.
- Most visitors will have a smartphone capable of loading a mobile web page.
- All host employees have a company email address where they can receive notifications.
- The first version of the system will be deployed for a single office branch.
- The development team has access to a Python and React development environment.

11.2 Constraints

- The project is being developed as part of an internship, so the timeline and team size are limited.
- The AI features (face authentication and pattern detection) are out of scope for the initial launch and will be planned for a future phase.
- Hardware integration (such as automatic door unlocking) is not part of this project.

12. Glossary

The following terms are used throughout this document. This glossary is included to make the document accessible to non-technical readers.

Term	Plain English Definition
QR Code	A square barcode that a camera can scan to instantly read information — in this case, a visitor's check-in details.
API	A connection point between two pieces of software. Think of it as a door that lets different parts of the system talk to each other.
RBAC (Role-Based Access Control)	A way of controlling what each user can see and do. A security guard can do different things than an admin — RBAC enforces that.
JWT (JSON Web Token)	A secure, digital 'ticket' that a user receives when they log in. It proves who they are to the system without needing to re-enter a password on every action.
HTTPS	The secure version of the standard web protocol. It means all data sent between the browser and the server is encrypted and cannot be easily intercepted.
AES-256 Encryption	A very strong method of scrambling data so that only someone with the right key can read it. The '256' refers to the length of that key — the longer, the harder to crack.
UAT (User Acceptance Testing)	The final phase of testing, done by actual end users, to confirm the system works the way they need it to before it goes live.
Dashboard	A screen that shows a live summary of important information — in this case, who is currently in the building, today's visitor count, and recent activity.
Offline Mode	A fallback mode that allows the system to keep working in a limited way even when there is no internet connection.

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